To reclaim the space in the table, you'll need to run VACUUUM FULL users; from psql. Running VACUUM FULL on a live system is not recommended because it will lock the table while it runs.

Indexes for the users table will also become bloated and should be rebuilt. This keeps them optimized for queries.

As you saw earlier, you may run REINDEX for a single Index, or for all Indexes on a table. Run this statement from psql to rebuild one index:

sql/reindex_concurrently.sql

```
REINDEX INDEX CONCURRENTLY index_users_on_email;
```

You've now seen how to use Vacuum, Analyze, and Reindex to help with your database maintenance. What other types of Index maintenance are there?

Removing Unused Indexes

The PostgreSQL Wiki has a dedicated page for *Index Maintenance*[12] worth reviewing.

Indexes are added to tables to be used by queries. PostgreSQL tracks their usage when a query execution plan performs an Index Scan.

To make sure Index Scans are tracked, confirm that the following values are set to on (they are on by default) by running these statements in psql.

```
SHOW track_activities;
SHOW track_counts;
```

Regularly search for indexes that aren't used by queries. These are called Unused Indexes and they should be removed. Indexes can consume a lot of space and they also add some latency to write operations as they're maintained.

- Unused Indexes may prevent Heap Only Tuple (HOT) Updates
- Unused Indexes unnecessarily increase the VACUUM run time for a table
- Unused Indexes unnecessarily increase query planning time
- Unused Indexes unnecessarily slow down backup and restore operations

Indexes can grow very large, even larger in size than the tables themselves. This can be very surprising at first. These very large Unused Indexes weigh down your system needlessly.

To remove an index safely on a live system, use the CONCURRENTLY keyword as you've done many times in earlier examples.

12. https://wiki.postgresql.org/wiki/Index_Maintenance